

Windows Kiosk Implementation Guide

Production hardening and deployment runbook for the public RVM and enterprise PecoDrop applications

Recommended platform: Windows 11 IoT Enterprise LTSC 2024

Applications: RVMDesktopApp and PecoDropDesktopApp

Workspace: D:\GIT-HUB\RVM-dash

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Internal deployment reference. Test every policy on a pilot kiosk before fleet rollout.

1. Executive recommendation

Keep the existing WPF applications on Windows. For new fixed-purpose machines use a genuine, licensed Windows 11 IoT Enterprise LTSC 2024 image. Do not use modified “lite Windows” distributions such as Tiny11 or Ghost Spectre; they can remove media, .NET, driver, servicing, or security components required by the kiosks.

Implementation order: production build → persistent logging → controlled startup → watchdog → recovery testing → keyboard lockdown → Shell Launcher → Unified Write Filter.

2. Keep the products separated

Application	Purpose and hardware identity	Deployment identity
RVMDesktopApp	General-public system: plastic, cans, UBC and rejects; rewards, coupons and public leaderboards.	Separate machine ID, configuration, serial port, ads, logs, credentials and data directories.
PecoDropDesktopApp	Enterprise dual-display system with QR scanner/keypad;  plastic,  cans/metal,  weighed paper; corporate workflow and ESG reporting.	Separate machine ID, organization binding, display layout, configuration, serial port, ads, logs and credentials.

3. Hardware and firmware baseline

- Modern Intel Core i3/i5, N100/N305, or equivalent.
- 8 GB RAM minimum; 16 GB preferred for dual-display PecoDrop.
- 256 GB reputable SSD, with at least 20% free space.
- TPM 2.0, wired Ethernet, industrial powered USB hub and short shielded USB cables.
- UPS, adequate cooling and BIOS temperature monitoring.

BIOS settings

- ☐ Restore after AC loss = Power On.
- ☐ Wake on LAN enabled when remote management requires it.
- ☐ Fast Boot disabled if it interferes with USB discovery.
- ☐ External USB boot disabled after deployment.
- ☐ BIOS administrator password securely recorded.

4. Windows accounts

Account	Rights	Use
KioskUser	Standard user	Automatic sign-in and kiosk application only.
KioskAdmin	Administrator	Technician maintenance; never used for routine kiosk operation.

Never place administrator passwords, database passwords or API secrets in launcher scripts. Use unique managed credentials and protect recovery information centrally.

5. Produce fixed Release packages

Build on a development or CI machine. Never install source code or compile Debug builds on a production kiosk.

```
dotnet publish .\RVMDesktopApp\RVMDesktopApp.csproj `
-c Release -r win-x64 --self-contained true `
-p:PublishSingleFile=false -o .\deploy\RVM

dotnet publish .\PecoDropDesktopApp\PecoDropDesktopApp.csproj `
-c Release -r win-x64 --self-contained true `
-p:PublishSingleFile=false -o .\deploy\PecoDrop
```

Install immutable application files under:

```
C:\Program Files\ISP\RVM\
C:\Program Files\ISP\PecoDrop\
```

Place mutable data under:

```
C:\ProgramData\ISP\RVM\
C:\ProgramData\ISP\PecoDrop\
```

6. Production launcher requirements

The repository's current launch scripts are useful for development, but a production launcher must:

1. Use one fixed executable and working directory.
2. Never search development folders or build missing executables.
3. Validate configuration, assets and available disk space.
4. Record startup, process ID, version, display information and failure codes.
5. Wait for readiness and confirm the required window count.
6. Exit with a defined status code so the watchdog can recover safely.

Current issue to remove: the RVM launcher can build a Debug executable when no binary is found. Production machines must fail safely instead of compiling or selecting another build.

7. Persistent logging

Create rolling logs in each application's own `ProgramData` directory. Retain about 30 days and enforce a total size limit.

- Application version, machine ID, boot/start/stop time and shutdown reason.
- Arduino discovery, port changes, reconnect attempts and calibration.
- API/database connectivity and offline queue count.
- Display enumeration, placement and video failures.
- Unhandled exceptions, watchdog decisions and crash dumps.

Never log passwords, tokens, wallet credentials or unnecessary personal information.

8. Independent watchdog

Use an independent signed watchdog plus a Scheduled Task in the interactive KioskUser session. A Windows service should monitor state, but it should not directly display the WPF UI from the service session.

```
Watchdog service → health decision
Scheduled task   → interactive launcher
Launcher        → RVMDesktopApp or PecoDropDesktopApp
```

Check every 15–30 seconds for process state, application heartbeat, responsiveness, memory, disk space and hardware connectivity. For PecoDrop also confirm both display windows.

Recovery sequence

1. Recheck after a short delay to avoid false alarms.
2. Request graceful shutdown.
3. If unresponsive, terminate only the kiosk process.
4. Restart the application.
5. After repeated failures, report a crash loop and reboot only at a configured threshold.

Use a limit such as five restarts in ten minutes. PecoDrop's Arduino host lease must close all mechanisms safely when application communication stops.

9. Startup and Shell Launcher

During initial deployment, use Task Scheduler:

- Trigger at KioskUser logon with a 15–30 second delay.
- Run only in the interactive user session.
- Restart on failure every minute, with a bounded retry count.
- Use a fixed working directory and production launcher.

After field testing, Windows IoT Shell Launcher can replace `Explorer.exe`. Before enabling it, document and test the technician recovery path and the method to restore Explorer.

10. Controlled Windows updates

- ☐ Keep security and quality updates enabled.
- ☐ Set active hours to cover the full operating day.
- ☐ Schedule installation/restart during a maintenance window.
- ☐ Suppress public update and restart notifications.
- ☐ Update one pilot kiosk first; deploy to the fleet after 3–7 stable days.
- ☐ Do not install untested GPU, USB or display drivers fleet-wide.

11. Power, notifications and performance

- ☐ Sleep, hibernation, display timeout and screen saver disabled.
- ☐ Notifications, consumer suggestions and Game Bar disabled for KioskUser.
- ☐ Microsoft Defender and Windows Firewall remain enabled.
- ☐ OEM trial antivirus and unnecessary startup utilities removed.
- ☐ UPS shutdown and recovery tested by physically removing mains power.

12. USB and Arduino stability

- Disable USB selective suspend and USB hub power-down.
- Identify Arduino hardware by VID/PID/serial where possible, not only by a fixed COM number.
- Use bounded reconnect attempts with increasing delay.
- Never let an admin window open a competing serial connection.
- Log arrival, disappearance and reassignment of the port.

PecoDrop safe reconnect order

```
USB detected → serial connection → HOST:ALIVE → HOST:START  
→ calibration request/retry → CALIBRATION:OK  
→ verified compartment state → intake permitted
```

Never treat startup METAL/PAPER full-bin readings as authoritative before calibration completes.

13. Display configuration

Public RVM

Fix the native resolution, scale, orientation, primary monitor and touchscreen mapping. Test the adaptive UI on 1080p and taller kiosk screens; avoid fixed heights that clip the live-session breakdown or leaderboard.

Enterprise PecoDrop

Test both dual-landscape arrangements:

- `0012`: hardware kiosk left, video signage right.
- `0021`: video signage left, hardware kiosk right.

Test boot with one display missing, HDMI hot-plug, monitor replacement, GPU restart and Windows restart. The machine-scoped SQL display setting remains authoritative; `config.txt` is the offline fallback.

14. Lockdown features

After stability testing, apply Windows IoT features in stages:

- **Keyboard Filter:** block public escape shortcuts such as Alt+F4, Alt+Tab, Windows+R and Ctrl+Shift+Esc.
- **Shell Launcher:** replace the normal Explorer shell.
- **Custom Logon and unbranded boot:** remove public Windows UI during startup.
- **Unified Write Filter (UWF):** protect the system volume only after persistent-data planning.

Recovery warning: never block technician access until a tested maintenance escape and rollback procedure exists.

15. Unified Write Filter design

Prefer a protected system/application volume and a persistent data volume:

```
C: Windows and application binaries – UWF protected  
D: Logs, database, offline queue, settings and downloads – persistent
```

Complete activation, drivers and initial updates before enabling UWF. Monitor overlay consumption. Ensure logs, mutable configuration, local database, offline transactions, advertisements, display preference and crash dumps persist across reboot.

16. Security

- ☐ BitLocker enabled and recovery keys centrally protected.
- ☐ Firewall denies unnecessary inbound traffic.
- ☐ Application packages signed and checksums recorded.
- ☐ Unique machine/API identity per kiosk.
- ☐ Production secrets not stored as plaintext in `config.txt`.
- ☐ USB storage restricted without blocking Arduino and scanner devices.
- ☐ Remote maintenance available only through a controlled VPN or management system.
- ☐ Kiosks communicate through the HTTPS API wherever possible rather than direct database exposure.

17. Health monitoring

Send application version, uptime, connectivity, calibration state, offline queue count, free disk, memory, display count and last restart reason to the Web Dashboard.

Raise alerts for:

- No heartbeat for five minutes.
- Repeated application restart or memory growth.
- Arduino disconnection or incomplete calibration.
- Disk below 15% free space or SSD health warning.
- Increasing offline queue.
- PecoDrop operating with fewer than two displays.
- Windows awaiting restart beyond the approved maintenance interval.

18. Pilot and rollout checklist

1. Prepare and update one reference PC.
2. Install drivers and the fixed Release package.
3. Create KioskUser and KioskAdmin.
4. Configure displays, touchscreen, scanner and Arduino.
5. Enable persistent logging, startup task and watchdog.
6. Test power loss, network loss, API/database outage and recovery.
7. Test USB unplug/replug and COM reassignment.
8. Test application crash and crash-loop handling.
9. Test Windows updates and driver rollback.
10. Enable Keyboard Filter, then Shell Launcher.
11. Enable UWF only after all earlier tests pass.
12. Run continuously for 7–14 days and review logs.
13. Pilot the public RVM first; then perform a longer dual-display PecoDrop pilot.

19. Recommended repository deliverables

```
deployment/  
  rvm/  
    Install-RVMKiosk.ps1  
    Uninstall-RVMKiosk.ps1  
    Start-RVMKiosk.ps1  
    HealthCheck-RVM.ps1  
  pecodrop/  
    Install-PecoDropKiosk.ps1  
    Uninstall-PecoDropKiosk.ps1  
    Start-PecoDropKiosk.ps1  
    HealthCheck-PecoDrop.ps1  
  common/  
    Configure-WindowsKiosk.ps1  
    Configure-Power.ps1  
    Configure-Updates.ps1  
    Configure-Logging.ps1  
    Rollback-KioskMode.ps1  
  docs/  
    RVM-Deployment.md  
    PecoDrop-Deployment.md  
    Technician-Recovery.md
```

20. Official references

- [Windows 11 IoT Enterprise LTSC 2024 lifecycle](#)
- [Shell Launcher overview](#)
- [Windows IoT device-lockdown features](#)
- [Microsoft kiosk and Assigned Access recommendations](#)

End of guide — RVM and PecoDrop configuration, data and hardware ownership must remain separate at every deployment stage.